practcal:2

```
import numpy as np
```

```
ary_1=np.array([1, 2,3,4,5,6,7])
```

```
print(ary_1)
```

```
[1 2 3 4 5 6 7]
```

```
type(ary_1)
```

```
numpy.ndarray
```

```
print(ary_1.shape)
```

```
(7,)
```

```
print(ary_1.ndim)
```

```
1
```

```
print(ary_1.size)
```

```
7
```

```
ary_2=np.array([[1,2,3,4,5,6,7,8,9],[11,12,13,14,15,16,17,18,19],[111,222,333,444,555,666,777,888,999]])
```

```
print(ary_2)
```

```
[[  1   2   3   4   5   6   7   8   9]
 [ 11  12  13  14  15  16  17  18  19]
 [111 222 333 444 555 666 777 888 999]]
```

```
print(ary_2.shape)
```

```
(3, 9)
```

```
print(ary_2.size)
```

```
27
```

```
print(ary_2.ndim)
```

```
2
```

```
ary_3=np.arange(1,29)
```

```
print(ary_3)
```

```
[ 1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24
 25 26 27 28]
```

```
ary_4=ary_3.reshape(2,2,7)
```

```
print(ary_4)
```

```
[[[ 1  2  3  4  5  6  7]
  [ 8  9 10 11 12 13 14]]

 [[15 16 17 18 19 20 21]
```

```
       [22 23 24 25 26 27 28]]]
```

```
np.ones([3,3])
```

```
array([[1., 1., 1.],
       [1., 1., 1.],
       [1., 1., 1.]])
```

```
np.zeros([3,3])
```

```
array([[0., 0., 0.],
       [0., 0., 0.],
       [0., 0., 0.]])
```

```
np.identity(3)
```

```
array([[1., 0., 0.],
       [0., 1., 0.],
       [0., 0., 1.]])
```

```
np.where(ary_3==28)
```

```
(array([27]),)
```

```
np.where(ary_4==28)
```

```
(array([1]), array([1]), array([6]))
```

```
np.where(ary_3>=15)
```

```
(array([14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27]),)
```

```
np.where(ary_3<=15)
```

```
(array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14]),)
```

```
np.where(ary_3)
```

```
(array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16,
        17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27]),)
```

```
ary_3>=13
```

```
array([False, False, False, False, False, False, False, False, False,
       False, False, False,  True,  True,  True,  True,  True,  True,
        True,  True,  True,  True,  True,  True,  True,  True,  True,
        True])
```

```
ary_3<=13
```

```
array([ True,  True,  True,  True,  True,  True,  True,  True,  True,
        True,  True,  True,  True, False, False, False, False, False,
       False, False, False, False, False, False, False, False, False,
       False])
```

```
ary_3[ary_3>15]
```

```
array([16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28])
```

Start coding or generate with AI.

```
ary_5=np.arange(1,100,4)
```

```
print(ary_5)
```

```
[ 1  5  9 13 17 21 25 29 33 37 41 45 49 53 57 61 65 69 73 77 81 85 89 93
 97]
```

```
ary_5.sum()
```

```
np.int64(1225)
```

```
ary_6=np.arange(1,21)
```

```
print(ary_6)
```

```
[ 1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20]
```

```
ary_7=ary_6.reshape(2,2,5)
```

```
print(ary_7)
```

```
[[[ 1  2  3  4  5]
  [ 6  7  8  9 10]]

 [[11 12 13 14 15]
  [16 17 18 19 20]]]
```

```
ary_7.sum()
```

```
np.int64(210)
```

```
ary_7.sum(axis=0)
```

```
array([[12, 14, 16, 18, 20],
       [22, 24, 26, 28, 30]])
```

```
ary_7.sum(axis=1)
```

```
array([[ 7,  9, 11, 13, 15],
       [27, 29, 31, 33, 35]])
```

```
arr_8=np.arange(1,10)
```

```
ary_9=arr_8.reshape(3,3)
```

```
print(ary_9)
```

```
[[1 2 3]
 [4 5 6]
 [7 8 9]]
```

```
ary_9[2,2]
```

```
np.int64(9)
```

```
ary_9[1]
```

```
array([4, 5, 6])
```

```
ary_9[: ,1]
```

```
array([2, 5, 8])
```

```
ary_9[1:2 , 0:1]
```

```
array([[4]])
```

```
ary_9[-1,-1]
```

```
np.int64(9)
```

```
ary_9.sum()
```

```
np.int64(45)
```

```
ary_9.sum(axis=0)
```

```
array([12, 15, 18])
```

```
ary_9.sum(axis=1)
```

```
array([ 6, 15, 24])
```

```
ary_9.max()
```

```
np.int64(9)
```

```
ary_9.min()
```

```
np.int64(1)
```

```
ary_9+2
```

```
array([[ 3,  4,  5],
       [ 6,  7,  8],
       [ 9, 10, 11]])
```

```
ary_9-3
```

```
array([[-2, -1,  0],
       [ 1,  2,  3],
       [ 4,  5,  6]])
```

```
ary_9.mean()
```

```
np.float64(5.0)
```

```
ary_9.mean(axis=0)
```

```
array([4., 5., 6.])
```

```
ary_9.mean(axis=1)
```

```
array([2., 5., 8.])
```

```
print(np.median(ary_9))
```

```
5.0
```

```
print(np.median(ary_9,axis=0))
```

```
[4. 5. 6.]
```

```
print(np.median(ary_9,axis=1))
```

```
[2. 5. 8.]
```

```
ary_9.var()
```

```
np.float64(6.666666666666667)
```

```
ary_9.std()
```

```
np.float64(2.581988897471611)
```